Lab 6: 3D Transformations

LAB SHEET

Theory: Write down the transformation matrix of 3D translation, rotation and scaling.

Lab works:

Question 1: Write a code in C to implement 3D translation.

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<math.h>
int maxx, maxy, midx, midy;
void axis()
{
	//getch();
	cleardevice();
	line(midx,0,midx,maxy);
	line(0,midy,maxx,midy);
}

int main()
{
	int x,y,z,o,x1,x2,y1,y2;
	int gd=DETECT,gm;
	detectgraph(&gd,&gm);
	initgraph(&gd,&gm,"");
	maxx=getmaxx();
```

```
	maxy=getmaxy();
	midx=maxx/2;
	midy=maxy/2;
	axis();
	bar3d(midx+50,midy-100,midx+60,midy-90,10,1);
	printf("Enter translation factor \n");
	scanf("%d%d",&x,&y);
	printf("After translation: Check the output screen");
	bar3d(midx+x+50,midy-(y+100),midx+x+60,midy-(y+90),10,1);
	getch();
	closegraph();
	return 0;
}
```

Question 2: Write a code in C to implement 3D scaling.

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<math.h>
int maxx,maxy,midx,midy;
void axis()
{
	cleardevice();
	line(midx,0,midx,maxy);
	line(0,midy,maxx,midy);
}
```

```
int main()
{
int x,y,z,o,x1,x2,y1,y2;
int gd=DETECT,gm;
detectgraph(&gd,&gm);
initgraph(&gd,&gm,"");
maxx=getmaxx();
maxy=getmaxy();
midx=maxx/2;
midy=maxy/2;
axis();
bar3d(midx+50,midy-100,midx+60,midy-90,5,1);
printf("Enter scaling factors \n");
scanf("%d%d%d", &x,&y,&z);
printf("After scaling");
bar3d(midx+(x*50),midy-(y*100),midx+(x*60),midy-(y*90),5*z,1);
getch();
closegraph();
return 0;
}
```

Question 3: Write a code in C to implement 3D rotation. (DO IT YOURSELF)

Conclusion

- What did u learn?